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BSc. in Textile Engineering

Project Report on

Comparing the Properties of Ring and Vortex Yarn made from
blend of Viscose, Recycled Cotton and Polyester

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Abstract

Sustainable development, environment protection, zero wastage are the top most concerns now a days. Almost half of the textiles used by human are made of cotton. But cotton production needs much more resources and it hampers food production. Cotton recycling requires less resources and it reduces wastage. Regenerated cotton has some superior qualities and it is also environment friendly than cotton. But recycled cotton and regenerated cotton lacks in some physical properties such as strength, elongation and lusture. So, we added regenerated polyester with that. As polyester is a synthetic fiber, it requires a long time to degrade in nature. Regenerating polyester saves environment and reduces wastage.

In this project, we blended recycled cotton, modal and recycled polyester at a ratio of 30:40:30. We made some other ratios but this blend gives the maximum mechanical properties. Then we produced 20 and 30 Ne yarn in both ring and air-vortex spinning machine and compared the properties of two yarns.

Acknowledgement

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1. Introduction

1.1. Introduction

According to their source of origin, fibers are classified into two groups: natural fiber and manmade fiber or synthetic fiber. Among the natural fibers, cotton is the most widely used fiber, with a variety of applications, due to its supreme and unique characteristics. With outstanding moisture absorption and thermal qualities associated with comfort, cotton is by far the most important natural fiber on the market.

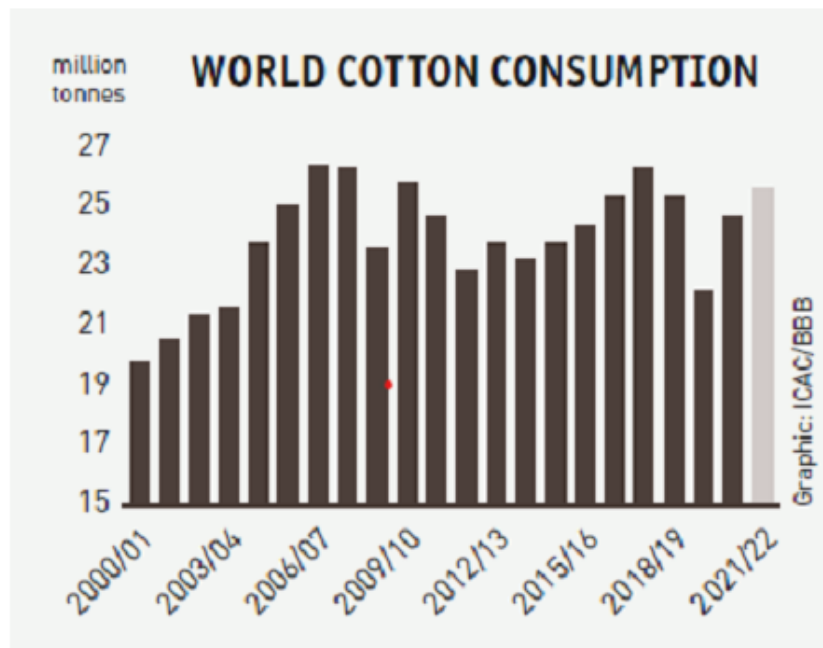


Figure 1: World cotton consumption

Compared to natural fiber, manmade fiber has many advantages in terms of quality and properties over natural fiber, which has driven the growth of demand for manmade fibers in the global textile market in the last few decades. From the manmade fiber group, polyethylene terephthalate (PET) is the most widely used synthetic polyester textile material for both home and industrial purposes, as the uses of polyester fiber can range from traditional textiles to technical textiles. PET takes the top spot as the polyester for fiber manufacturing, not only because of its good end-use qualities and low cost of

manufacture, but also because of its ease of physical and chemical modification, which can suppress undesirable properties while improving positive ones. Molecular structure and the presence of aliphatic and aromatic components in macromolecular chains give these fibers their distinct characteristics.

In 2016, the total amount of polyester fiber, or PET fiber produced exceeded 50 million tons, and in 2020 it exceeded 56 million with a growth rate significantly greater than any other natural or synthetic fiber. It is also estimated that in 2030, the production of polyester fiber will be almost 70 million tons, thus continuing its dominance over other fibers. The following are some reasons why PET fiber is so popular: low cost, easy processing, wash-and-wear qualities, high modulus coupled with outstanding modulus retention, recycling, reuse, adaptable performance, and blendability with cotton and other natural fibers.

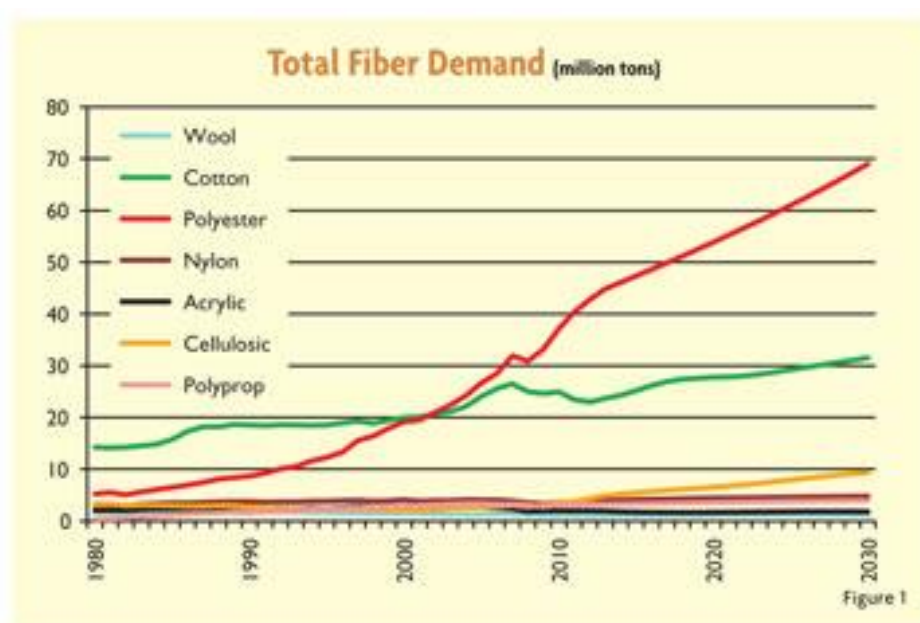


Figure 2: Change in demand of different fibers

Modal is a generic name for a modified rayon fiber that has high tenacity and high wet modulus. Modal fibers were initially developed in the 1930s for industrial uses in tires, conveyor belts and hose pipes. Additional developments in Japan in 1951 by S. Tachikawa led to the production of modal fiber with high wet modulus called polynosic. Modal fibers are dimensionally stable and do not shrink or get pulled out of shape when wet like many rayons. They are also wear resistant and strong while maintaining a soft, silky feel.

1.2. Background of the study

Among the basic needs of human being clothing is one of them. With the rapid growth of population demand for apparels and other clothing items also grew enormously. This demand puts a huge impact on cotton cultivation. Numerous chemicals and pesticides are used in cotton cultivation to minimize the harvesting period and maximize the production. But this increase uses of chemicals imparts a negative impact on environment and make the soil unfertile. On the other hand, rapid change in fashion increase clothing wastage at an alarming rate, as the decomposition of fabrics take a good amount of time. To handle this concerning problem recycling the cotton fibre from waste materials such as old apparels, fabric cutting wastage and other process wastage would be the best solution. Using recycle cotton can cause a great deal both economically and environmentally. Almost 60-70% cost in yarn manufacturing is associated with raw material. Using recycle cotton can drastically reduce this cost and increase profit margin for manufacturer. The main obstacle of using recycle cotton instead of conventional fibres is it's strength, which is 15 - 25 gram per denier on average. While virgin cotton strength can rise up to 40-45 gram per denier. To overcome this issue recycle cotton must be blended with other fibres. Often synthetic fibres have high strength but it takes a lot of time to decompose. So, the uses of synthetic fibres must be limited. To do so instead of using conventional two fibre blend the no of fibres can be increased in the blend. We blended three fibers (recycle cotton, viscous and polyester) with different ratio and compared the performance with conventional yarn made of virgin cotton.

1.3. Objectives of fiber recycling

Farming cotton is extensively resource-intensive. It requires pesticides and fertilizers which causes river pollution and damages farmers health. It takes a lot of water to produce textile, plus land to grow cotton and other fibers. To make a single cotton t-shirt, 2,700 liters of fresh water are required according to estimates, enough to meet one person's drinking needs for 2.5 years. To reduce the environmental impact, burden on natural resources and social problems associated with cotton, the idea of regenerating cotton and recycling cotton come out.

Polyester is a synthetic fiber. It can't be degraded in nature by micro-organisms. A single laundry load of polyester clothes can discharge 700,000 microplastic fibers that can end

up in the food chain. Washing synthetic products leads to the accumulation of more than half a million tons of microplastics on the bottom of the oceans every year. In addition to this global problem, the pollution generated by garment production has a devastating impact on the health of local people, animals and ecosystems where the factories are located.

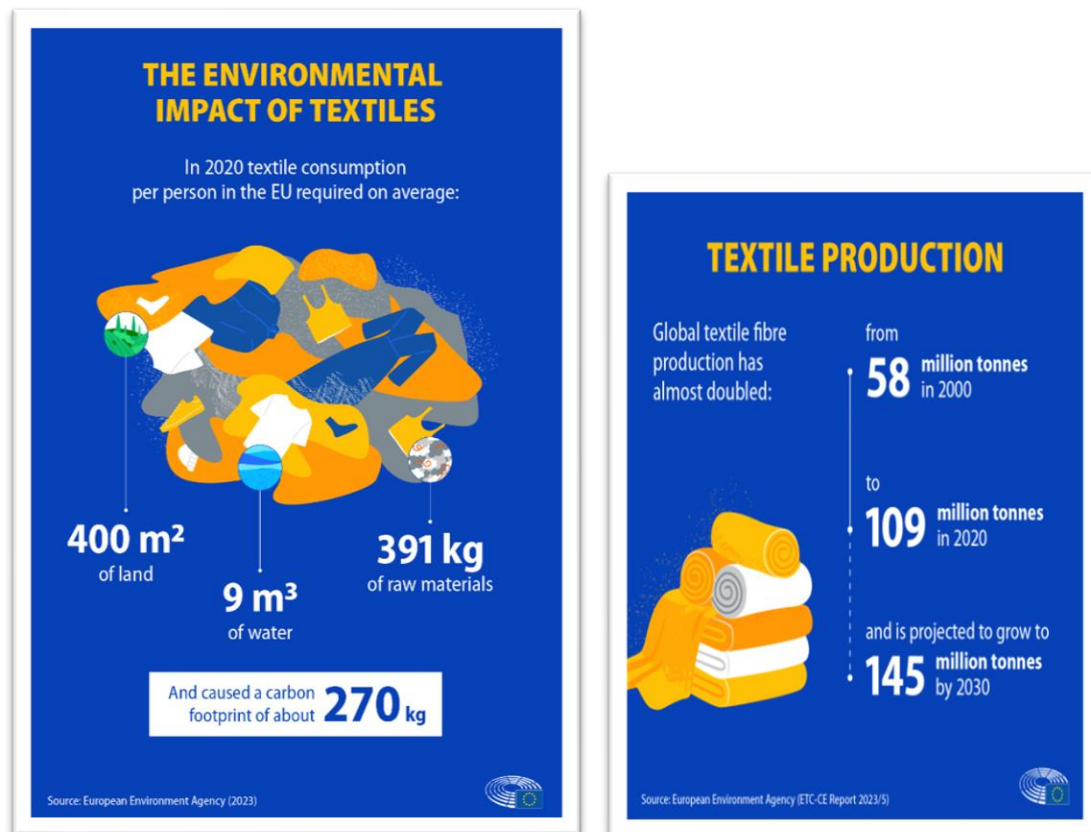


Figure 3: Effect of per person consumption of textiles

The way people get rid of unwanted clothes has also changed, with items being thrown away rather than donated. Less than half of used clothes are collected for reuse or recycling, and only 1% of used clothes are recycled into new clothes, since technologies that would enable clothes to be recycled into virgin fibers are only now starting to emerge.

2. Literature review

2.1. Cotton

2.1.1. Introduction

Cotton is a natural staple fiber. It is soft, fluffy. It is found in seed pods of the cotton plant. It is a shrub type perennial plant. A plant poses 100 cotton balls. A ball contains 27-45 seeds. Total amount of fiber in a ball is about 500000. It grows well in tropical or sub-tropical climates. Sunshine, heavy rainfall and higher humidity is required for its well growth.

2.1.2. Properties of cotton

2.1.2.1. Physical properties

Length – 10-65 mm

Diameter – 12-20 micrometers

Strength – 24-32 gm/tex

Tenacity – 26.5-44.1 cN/tex

Fineness – 3-6 Mic

Crystallinity – 70%

Elastic properties – rigid fiber

Specific gravity – 1.48-1.51 gm/cm³

Maturity ratio – 0.7-0.9

Resiliency – low

Dimensional stability – good

Moisture regain – 8.5%

Nature – hygroscopic

2.1.2.2. Thermal properties

Heat conductivity – better conductor than wool/silk

Luster – no luster

Shrinkage – does not shrink

Drape ability – not good

Residue – pale grey, powder ash

Combustion – quick, bright with afterglow

2.1.2.3. Chemical properties

Effects of alkali – resistant

Effects of acid – weakened & destroyed by acids

Effect of bleaches – no effects

Effect of sunlight & weather – fiber discoloration

Affinity to dyes – easy to dye & print

Effect of perspiration – discolours the fiber

2.1.2.4. Biological properties

Micro-organism – damage the cotton

Insects – no effect or damage

2.1.2.5. Electrical properties

No flow of electricity (Tensile properties of cotton fibers, 2009)

2.1.3. Environmental impacts of cotton

Carbon emissions from cotton production amount to around 220 million metric tons yearly. Regular cotton use a considerable amount of synthetic fertilizers, which release nitrous oxide into the atmosphere. Nitrous oxide is a powerful greenhouse gas that is 310 times more potent than carbon dioxide. Cotton is water-intensive as a crop and in manufacturing. About 2,700 liters of water are needed to produce a single cotton t-shirt. It

takes more than 20,000 liters of water to produce 1 kg of cotton. Cotton farms are usually monocultural. And as with monocultural systems sustained for decades, the soil quality suffers degradation. Cotton uses 24% and 11% of the world's insecticides and pesticides, respectively. It also uses 4% of the world's artificial phosphorus and nitrogen fertilizers which threaten human health, wildlife, water, and soil. Pollution from processing cotton into clothes or fabric for industrial use can affect humans as much as it affects wildlife and the environment.

2.1.4. Cotton recycling process

1. Fabric storing after checking. Ensuring there is no lycra or polyester. If fabric contains lycra or polyester, it will remain undyed and specks or moire effect will be visible after dyeing.



Figure 4. Wastage fabric go-down

2. Fabric is cut into shorter length by cutter machine which is suitable for opening.



Figure 5. Cutter machine

3. The recycling machine opens the fabric into fibers and individualizes the fibers. The fiber length remains close to 19 inch.



Figure 6. Recycling machine part-1

4. A mixer mixes the fibers so that shade change could not occur.

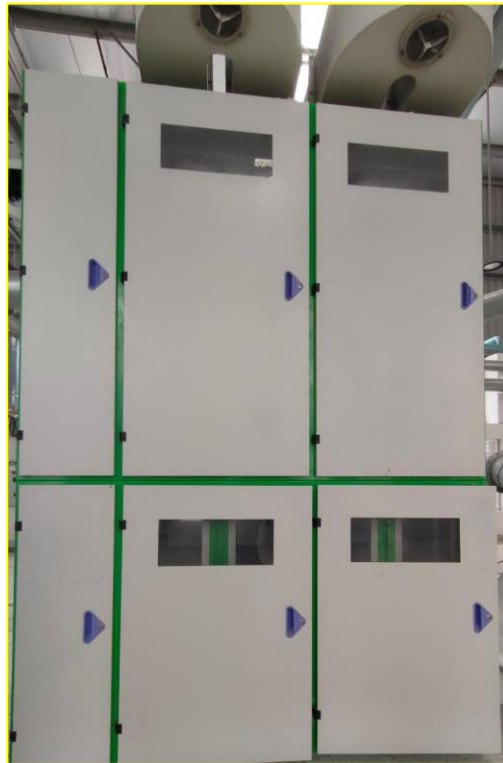


Figure 7. Multimixer

5. Recycling machine part-2 opens the fibers again so that fibers become straight and unentangled.



Figure 8. Recycling machine part-2

6. Finally fibers are pressed into bales and sent to blowroom.



Figure 9. Bale press

2.2. Recycled polyester (Coolmax)

There are 2 ways to recycle polyester: For mechanical recycling, plastic is melted to make new yarn. This process can only be done a few times before the fiber loses its quality. Chemical recycling involves breaking down the plastic molecules and reforming them into yarn. The main objective of recycling polyester is Keeping plastics from going to landfill and the ocean. Recycled polyester is just as good as virgin polyester but takes less resources to make. It Reduces CO₂ emissions by 25% vs. virgin counterparts. Choosing fibers made from textile waste helps those with a circular economy goal work towards achieving it. But recycling plastic has its limitations - Many garments are not made from polyester alone, but rather a blend of polyester and other materials. In that case, it is more difficult, if not impossible, to recycle them. Besides, the process of recycling PET impacts the environment too. Recycled polyester releases microplastics more than virgin polyester.

2.2.1. Application area of Coolmax

Suitable for a broad range of garment types that meet consumer needs. Such as-

- Activewear
- Bedding
- Denim
- Outerwear
- Ready-to-wear
- Seamless
- Socks
- Underwear
- Workwear

2.2.2. Need for recycled polyester

Polyester fiber is basically prepared from petroleum which is a high demand nonrenewable fuel source. The extraction of petroleum from crude oil creates a negative impact on the wellness of the ecosystem. The pollution caused due to plastic creates an extensive dispute among mankind; furthermore, the toxic and non-degradable nature of the plastic materials complicates the ecosystem. The extended utilization of polyester has increased the emission of polyester waste into the environment. These materials do not pose any direct damage to the living organisms and the ecosystem, but their nonbiodegradable nature remains as one of the biggest threat worldwide resulting in the elevation of polyester solid waste.

2.2.3. Process of polyester recycling



Figure 10: Polyester recycling flowchart

(A study on the suitability of which yarn number to use for recycle polyester fiber,

16 Jan 2019.)

2.2.4. Properties of Recycled Polyester

Table 1: Properties of Recycled Polyester

Properties	Values
Tensile Strength	55-65 MPa
Tensile Modulus	3600-2800 MPa
Flexural Strength	182-192 MPa
Elongation at Break	1.9-2.3%
Heat Deflection Temperature	63-71 °C
Water Absorption after 24 hrs at 23°C	15-19 mg
Density	1.10-1.11 gm/cm ³

(Erosive Wear Behavior of Jute Fiber / Polyester resin with Biowaste Materials, 2020)

2.2.4.1. Physical Properties

The recycled polyesters have remarkable physical properties such as bending length, moisture regain, abrasion resistance, tenacity and crease recovery angle. The recycled polyesters possess enhanced moisture regaining capacity than the virgin polyesters because of the significant molecular arrangement. In addition, the higher tenacity also

enhances the mechanical strength of the recycled polyester. The virgin polyesters seemed to possess higher molecular weights, higher viscosity and less melt flow index whereas the recycled polyesters possess less molecular weight, less viscosity because of the impact of higher temperature and shear caused during the recycling process of the PET.

2.2.4.2. Mechanical Properties

The mechanical properties of the polyesters are estimated in the form of tensile strength, flexibility, randomness and defects present in the yarn and fabric. The studies concluded that the recycled polyesters possess a compressive strength of about 73.7 MPa, Tensile strength of about 7.85 MPa, Flexural strength of about 22.4 MPa and elasticity of about 27.9 MPa at a time period of 7 days.

2.2.4.3. Environmental properties

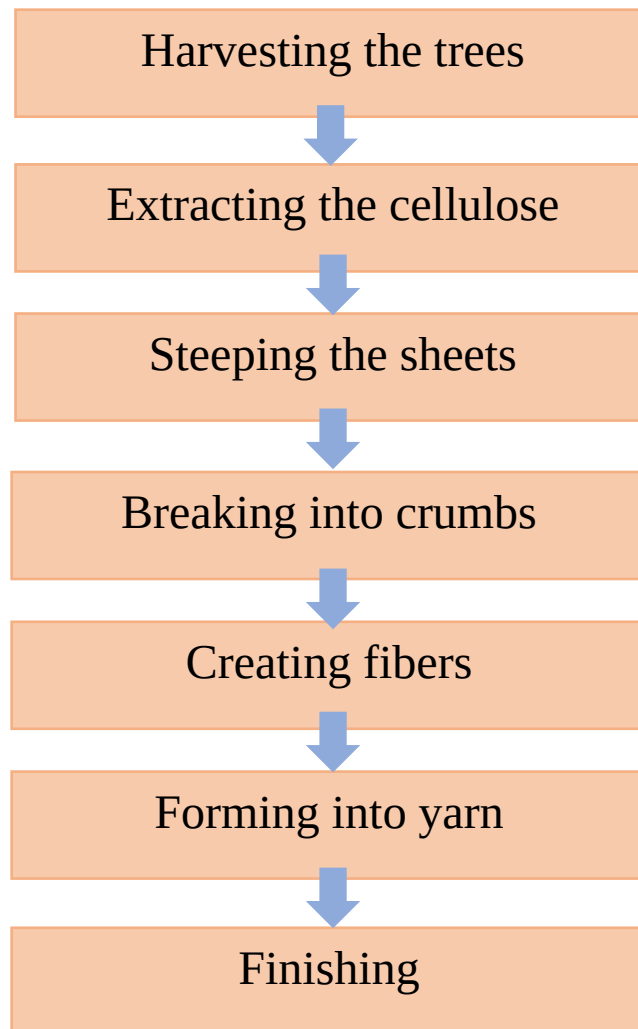
The degradation of plastic materials results in the release of several polyesters such as polylactic acid, poly-3-hydroxyalkanoate, polybutylene succinate and polycaprolactone. Many of the polyesters found in our environment are capable of degrading under the presence of some enzymes produced by microorganisms. The enzymatic degradation of polyesters occurs through chemical processes such as hydrolysis, de-polymerization and conversion into CO₂ and H₂O. One of the main limitations in the enzymatic degradation of the polyesters is the less occurrence of these polyester degrading bacteria (Recycled Polyester).

2.3. Modal

Modal rayon was first developed in 1951 in Japan, and in many ways, this type of rayon is nearly identical to viscose rayon. However, a simpler process is used to make modal rayon that doesn't produce as much waste, and many consumers note a significant difference in the feel of modal rayon as compared to viscose rayon. Lenzing started selling modal fibers in 1964. In 1977, Lenzing started using an environmentally friendly bleaching method for pulp for their cellulosic fibers. Modal's distinguishing characteristics are its high wet strength and its extra softness. It is sometimes referred to as "soft as a feather" and the "softest fiber in the world". (A Study on Modal Fibre Based on the Absorption Characteristics, July 29, 2015)

Modal is produced from wood of FSC (Forest Stewardship Council) certified forests. It is a regenerated fiber which uses less energy, less water and makes less CO₂ emission.

2.3.1. Production process



The modal fabric production process begins with the harvesting of trees to be rendered into cellulose. These trees are broken down into chips that are approximately the size of postage stamps, and they are transferred to the manufacturing floor. These chips are then purified to extract their cellulose content, and the leftover tree products are discarded.

Next, this extracted cellulose is formed into sheets, and these sheets are then immersed or "steeped" in vats of sodium hydroxide, which is also called caustic soda. It's important to note that far lesser concentrations of sodium hydroxide are used to create modal rayon than are used to create viscose rayon, which results in the production of less toxic waste. After steeping is complete, the sheets are broken down into crumbs. Unlike the case in the

production of viscose rayon, these crumbs are not aged, and they are instead directly immersed in carbon disulfide, which transforms the white crumbs into an orange substance called sodium cellulose xanthate. It is at this point that the tree-derived cellulose can no longer be considered a purely organic substance. This cellulose xanthate is then immersed in caustic soda once more to create a syrupy solution. The modal fabric production process again bypasses the aging process that usually occurs at this point, and this syrupy substance is immediately forced through a spinneret to create fibers.

These fibers are then immersed in sulfuric acid, stretched, and formed into yarn. The resulting yarn is then washed, bleached, rinsed, dried, and loaded onto spools. At this point, the modal yarn is ready to be subjected to pre-shrinking, fire resistance, wrinkle resistance, or a number of other treatments that may be applied before it is woven into a fabric.

2.4. Properties of fibers

Table 2: Properties of cotton, polyester and modal

Properties	Cotton	Polyester	Modal
Tenacity	3.00-4.90 gm/den	High tenacity filament = 7-8 gm/den Normal tenacity filament = 4.5-5.0 gm/den Staple fiber = 3.5-4.0 gm/den	Dry: 3.2-4.9 gm/den Wet: 1.8-2.6 gm/den
Elongation at Break	5-7%	20-30%	20-30%
Fiber Length	16-50 Mm	Filament	Filament
Moisture Regain	8.50%	0.40%	11%
Resiliency	Low	High	Better than viscose
Fineness	1.1-2.3 dtex	0.9-1.5 dtex	1.1-1.5 dtex
Effect Of heat	Decomposition starts at the temperature of about 149°C Rapid deterioration occurs at 246°C	Softening temperature: 230-245°C Melting temperature: 250-268°C	Decomposes at about 235°C

		Sticking temperature: 230-240°C Ironing temperature: 135°C Glass Transition temperature: 70°C	
Effect of sunlight	Loss of strength and damaged by UV	Have good resistance against sunlight but slowly becomes weaker on long time exposure to sunlight	Loss of strength and damaged by UV
Luster	Dull fiber	Highly Bright	Bright
Effect of acid	Dissolves in hot dilute or cold concentrated sulphuric acid	Highly resistant to mineral and organic acids. Weak acids cannot affect them even on boiling. Strong acids like sulphuric acid can affect them only after boiling for a long time.	Degraded by strong acid
Effect of alkali	Resistant to strong alkalis (e.g. NaOH)	Highly resistant to alkalis, only strong hot alkalis result in slow thinning of the filament diameter.	Resistant to alkali
Effect of bleaches	Not affected by oxidizing and reducing agents	Not affected by oxidizing and reducing agents	Less resistant to bleach than cotton
Dye Ability	Reactive dye or vat dye is mostly used for coloration	Disperse dyes and only some pigments can be used for coloration	Reactive dye is used for dying

(Study of mechanical and comfort properties of modal with cotton and regenerated fibers blended woven fabrics, February 2018)

2.5. Why blend of these fibers?

If we analyze the properties of the fibers, we see that polyester has the best mechanical properties. But it is less absorbent, more crystalline, less comfortable in nature. But aesthetic property of polyester is very good as it is highly lustrous. On the other hand, cotton has average mechanical properties. It is 65-70% amorphous, so very moisture absorbent and comfortable. Modal is a regenerated cellulosic fiber. It is mostly similar to cotton with some little differences. Mechanical properties of Modal are slightly weaker than cotton but it is softer, more lustrous, more absorbent and it can be produced at any

time, at any quantity as it is a regenerated fiber. Combination of these three fibers makes a unique high quality yarn which can be used extensively in some specific purposes such as in sportswear, in formal dresses.

3. Materials and Methodology

3.1. Materials

- Recycled cotton
- Recycled polyester (Coolmax)
- Modal

3.1.1. Recycled cotton

Source: DBL cotton recycling unit. Pre-consumed fabric waste and hard waste were used as raw material

Quality reports:

Table 3: HVI report of recycled cotton

SCI	109
Mst [%]	6.5
Mic	4.38
Mat	0.86
UHML [mm]	27.11
UI [%]	80.0
SF [%]	12.2
Str [g/tex]	28.9
Elg [%]	6.3
Rd	70.2
+b	10.8
CGrd	33-2

TrCnt	30
TrAr [%]	0.30
TrID LeafGrd	3
Amt	546

Table 4: AFIS report of recycled cotton

Total Nep Cnt [Cnt/g]	237
Total Nep Mean Size [µm]	668
Fiber Nep Cnt [Cnt/g]	225
FibNep Mean Size [µm]	656
SCNep Count [Cnt/g]	12
SCNep Mean Size [µm]	879
L(w) [mm]	25.8
L(w) CV%	33.3
SFC(w) %<12.7 mm	7.3
UQL(w) [mm]	31.7

3.1.2. Recycled polyester (coolmax)

It is produced in industry in a controlled manner. Thus, quality variation is not occurred. So, HVI and AFIS tests are not done. Properties of Coolmax are written in its cover.

Length: 38mm

Strength: 40 g/tex

Count: 1.5 den

3.1.3. Modal

It is produced in industry in a controlled manner. Thus, quality variation is not occurred. So, HVI and AFIS tests are not done. Properties of Modal are written in its cover.

Length: 38mm

Strength: 35 g/tex

Count: 1.5 den

3.2. Methods

Our aim is to make two ring yarn and two vortex yarn of 20 Ne and 30 Ne respectively and compare their properties. We are mainly focusing on sustainability while ensuring proper quality. Detailed specifications of our required yarn are given below:

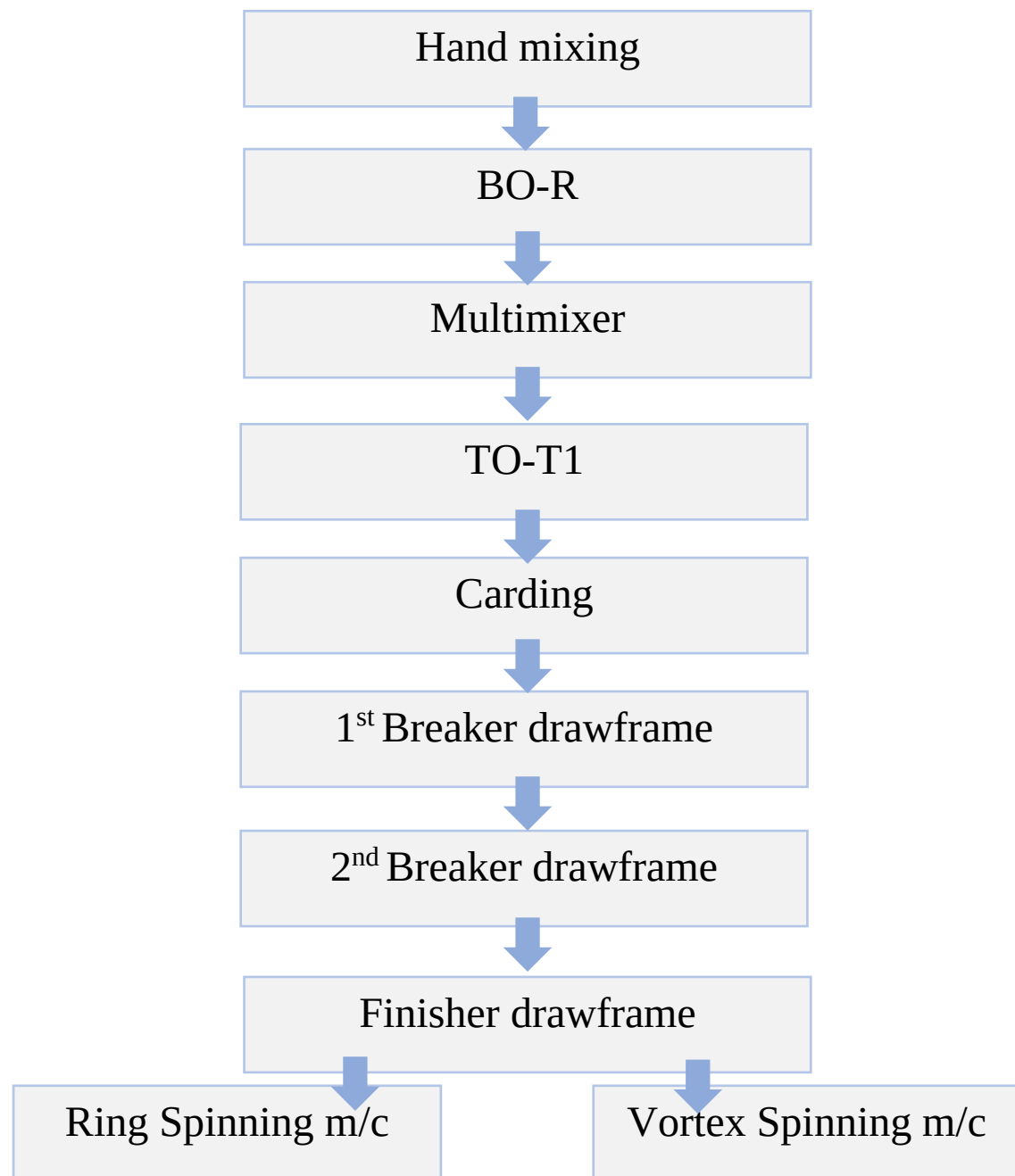
Count: 20/30 Ne

TPI: 22-27

Tenacity: 2200-2500 cN/tex

Color: White

3.2.1. Process flowchart



4. Sample Manufacturing

4.1. Machineries used

4.1.1. Waster Opener

Brand: Trutzschler

Model: BO-R

Input: Recycled cotton fiber

Output: Dust and trash free recycled cotton fiber



Figure 11: Waster opener BO-R

Compressed fiber bales are fed into it. It opens the bales into fiber flocks. It has beater in it which separates dust and trash. Impurities are removed by pneumatic system. It is an unavoidable machine as opening of fibers is the precondition of further processing.

4.1.2. Multimixer

Brand: Trutzschler

Model: MX- U(10)

Input: Recycled cotton, Recycled polyester (coolmax), Modal

Output: Homogeneously mixed fibers



Figure 12: Multimixer MX-U

Multimixer: It has 12 compartments in it. Each compartment can be filled individually. There is a weighing device in each compartment. Fibers can be mixed by certain percentage in this machine. We fed cotton, polyester and modal in separate compartments and blended by 30: 30:40 Ratio.

4.1.3. Opener

Brand: Trutzschler

Model: TO-T1

Input: Mixed fibers

Output: Card mat



Figure 13: Opener TO-T1

Opener: Opener opens the fiber flocks and removes dust, trash and impurities. It has toothed brush rollers which individualizes the fibers. It reduces pressure on card and increases quality. It produces card mat which is fed automatically to carding machine.

4.1.4. Carding

Brand: Trutzschler

Model: TC-11

Input: Card mat

Output: Card sliver (0.1 Ne)



Figure 14: Card TC-11

Carding machine is called the heart of the spinning system. It removes neps which is mostly generated in blowroom. It individualizes the fibers and removes dust, trash and short fibers. It produces card sliver.

4.1.5. Breaker drawframe

Brand: Rieter

Model: SB D15

Input: Card sliver (0.1 Ne)

Output: Breaker drawn sliver (0.1 Ne)



Figure 15: Breaker drawframe SB D-15

Breaker drawframe is used for reducing unevenness and parallelizing fibers in the sliver. We used 6:1 doubling in making this yarn. So thick thin place in the sliver is reduced by levelling effect. Then we drafted its for 6 times, so fiber became more parallel. If breaker drawframe was not used, there would be more unevenness in the yarn. We used two passages of breaker draw frame

4.1.6. Finisher drawframe

Brand: Rieter

Model: RSB D35

Input: Breaker drawn sliver (0.1 Ne)

Output: Finisher drawn sliver (0.1 Ne)



Figure 16: Finisher drawframe RSB D-35

Finisher drawframe has an auto-leveler in it. Auto-leveler detects thick/thin place and increase decreases roller speed accordingly. Thus, it reduces unevenness and produces a uniform sliver. It also doubles 6 slivers to produce 1 sliver.

4.1.7. Simplex

Brand: Toyota

Model: FL100

Input: Finisher drawn sliver (0.1 Ne)

Output: Roving (1.1 Ne)



Figure 17: Simplex machine Toyota FL100

Simplex machine produces roving from finisher drawn sliver. It is mostly used for giving draft. Finisher drawn sliver hank is 0.1 Ne and yarn count 30 Ne. $30/0.1=300$ draft is to be given to produce yarn from sliver. It is impossible to give 300 draft in Ring frame. Again, drawn sliver is kept in drum. Ring frame contains 1008 spindles. It is impossible to accommodate 1008 drums in front of ring frame. So, it is necessary to produce a suitable package for feeding ring frame. Simplex produces Roving which fulfill all the needs.

4.1.8. Ring Frame

Brand: Rieter

Model: MP1N

Input: Roving (1.1 Ne)

Output: Ring yarn (20/30 Ne)



Figure 18: Ring Frame MP1N

Key components of the ring frame include spindles, a drafting system, and the ring and traveler mechanism. The spindles rotate at high speeds, while the drafting system ensures that fibers are aligned and drawn out to the desired thickness. The ring and traveler work together to add twist to the fibers as they are spun, which is essential for creating durable yarn.

4.1.8.1. Air-Vortex

Brand: Murata

Input: Roving (1.1 Ne)

Output: Vortex yarn (20/30 Ne)



Figure 19: Murata vortex spinning (MVS)

The vortex spinning machine represents a significant advancement in the textile industry, particularly in the spinning of yarns. Unlike traditional spinning methods, which rely on twisting fibers together to form yarn, vortex spinning utilizes a unique process that involves a high-speed air vortex to entangle fibers. This innovative approach not only enhances the efficiency of yarn production but also improves the quality and characteristics of the final product.

At its core, the vortex spinning machine operates by drawing fibers into a spinning chamber where they are subjected to a rapidly rotating air vortex. This vortex creates a dynamic environment that allows for the fibers to be twisted and intertwined without the need for extensive mechanical twisting. As a result, the yarn produced is often stronger and more uniform than that made by conventional methods.

One of the key benefits of vortex spinning is its ability to handle a wide range of fiber types, including cotton, polyester, and blends. This versatility makes it suitable for various applications, from casual apparel to technical textiles used in specialized

industries. Additionally, the process generates less waste compared to traditional spinning techniques, contributing to more sustainable production practices.

Furthermore, vortex-spun yarns exhibit unique properties, such as improved softness, reduced pilling, and enhanced moisture management. These attributes make them increasingly popular in the fashion and home textiles markets.

4.2. Process parameters

4.2.1. Carding

Grain/yd	80
Speed	45 Kg/h
Flat speed	320 mm/min
Cylinder speed	320 mm/min
Taker in lever	4 (1-15)
Gauge	Normal

4.2.2. Breaker Drawframe (1st and 2nd)

Grain/yd	80
Doubling	6
Speed	700 m/min
Back zone gauge	42 mm
Front zone gauge	47 mm

4.2.3. Finisher Drawframe

Grain/yd	60
Doubling	6
Speed	550 m/min
Back zone gauge	42 mm
Front zone gauge	47 mm

4.2.4. Simplex

Roving hank	1.2 Ne
Draft	12
Speed	600 m/min
Back zone gauge	45 mm
Front zone gauge	49 mm

4.2.5. Ring Frame (20 Ne)

Total draft	18.18
Middle zone draft	13.97
Back zone draft	1.3
Condenser	Yellow
Spacer	2.4
Spindle	D1
Speed	15000 rpm
Bobbin color	Green

4.2.6. Ring Frame (30 Ne)

Total draft	25
Middle zone draft	19.23
Back zone draft	1.3
Condenser	Yellow
Spacer	2.4
Spindle	D1
Speed	15000 rpm
Bobbin color	Green

4.2.7. Air-Vortex (20 Ne)

MFs	Without
TDR	141
MDR	35
BDR	2.7
Feed ratio	0.99
Tension	140
Condenser	Cream
Spacer	2.4
Spindle	C1
Speed	420
Wax	With wax
Nozzle pressure	0.52

4.2.8. Air-Vortex (30 Ne)

MFs	Without
TDR	212
MDR	35
BDR	3.1
Feed ratio	0.99
Tension	120
Condenser	Gray
Spacer	2.4
Spindle	M1
Speed	420
Wax	With wax
Nozzle pressure	0.52

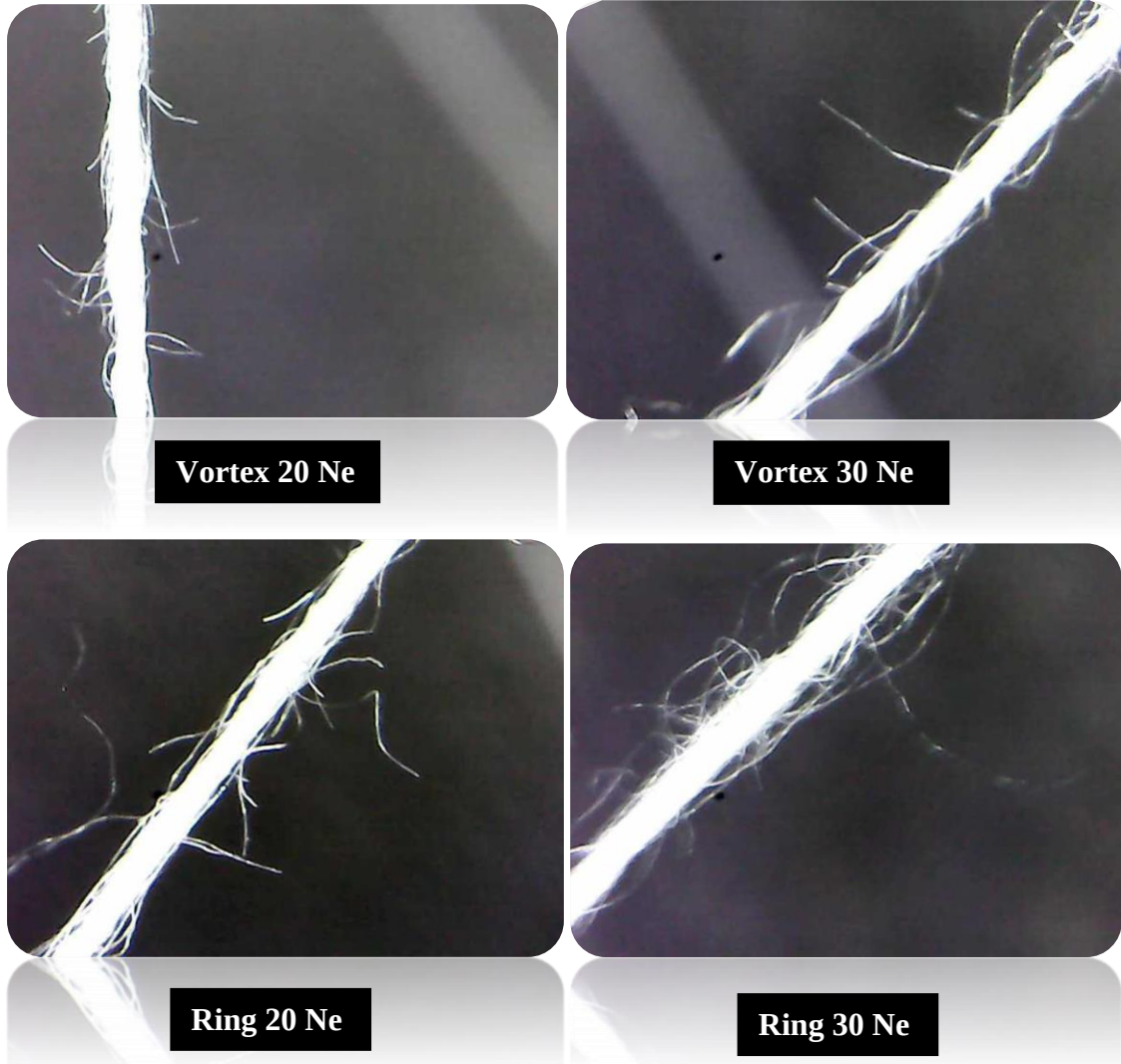
4.2.9. USTER AFIS Reports of intermediate products

Table 5: Properties of slivers

	Card mat	Card sliver	1st Breaker drawn sliver	2nd Breaker drawn sliver	Finisher drawn sliver
Total Nep Cnt [Cnt/g]	231	34	31	27	25
Total Nep Mean Size [µm]	660	559	564	566	567
Fiber Nep Cnt [Cnt/g]	218	34	31	26	25
FibNep Mean Size [µm]	628	559	564	569	567
SCNep Count [Cnt/g]	12	0	0	1	0
SCNep Mean Size [µm]	1248	0	0	475	0
L(w) [mm]	32.5	33.2	33.2	33.1	34.9
L(w) CV%	26.9	24	24	24.6	23.2
SFC(w) %<12.7 mm	3.3	1.9	2.2	2.1	1.5
UQL(w) [mm]	38.2	37.8	38.0	37.9	39.3
L(n) [mm]	27.8	29.7	29.7	29.4	31.6
L(n) CV%	41	34.1	34.7	35.3	32.5
SFC(n) %<12.7 mm	12.9	7.6	8.6	8.5	6.2
5% L(n) [mm]	40.8	40.6	40.7	40.7	42.2
Fineness [mtex]	165	160	163	160	165
Maturity Ratio [mat1]	1.02	1.03	1.04	1.03	1.06
IFC [%]	3.7	4.0	3.7	4.2	3.4

5. Result and discussion

5.1. Microscopic view of yarn



Above pictures shows the microscopic view of tri blend (Recycled cotton 30%, Modal 40%, Recycled polyester 30%) yarn produced in both ring and vortex m/c. Yarn produced in ring shows less hairiness, high uniformity and compactness.

5.2. Actual count of yarn

Type	Nominal count	Actual count
Ring	20 Ne	19.4 Ne
	30 Ne	29.25 Ne
Air-Vortex	20 Ne	19.51 Ne
	30 Ne	29.72 Ne

Due to scope of different errors (manmade and machine made) it is not possible to obtain target count 100% accurately. Slight variation occurs in yarn count. From the above data it is clearly visible that this variation is very minimum in vortex yarn compared to ring yarn.

5.3. USTER IPI & SPECT Report

Count	U%	CV _m %	H	Thin(-50%)/ km	Thick(+50%)/ km	Neps(+200%)/ km
20 (Ring)	7.63	9.64	4.96	0	4	7
30 (Ring)	9.54	12.08	3.78	1	26	46
20 (Vortex)	9.35	11.79	4.64	0	8	12
30 (Vortex)	11.2	14.11	4.25	16	34	48

U%:

It indicates the percentage of unevenness in a sample of yarn. A lower U% value signifies better quality, while a higher value indicates more defects. From the table U% of vortex yarn is 1.72% higher than ring yarn in 20 Ne yarn. And in 30 Ne it is 1.66% higher. But in 100% virgin cotton U% is slightly lesser.

Hairiness:

Hairiness refers to the presence of protruding fibers on the surface of yarn. In case of 20 Ne vortex yarn shows better hairiness property but in case of 30 Ne ring yarn shows better property. Comparing to 100% virgin cotton all of them shows almost similar hairiness property.

Neps/km:

If the thickness of any portion of yarn is 200% higher than average then the portion is considered as nep. It is measured as no of neos per km. From the above data ring yarn has less neps/km than vortex yarn. Comparing this with 100% virgin cotton it shows 10-15 more neps/km.

Thin (-50%):

This refers to the area of the yarn where yarn diameter is half of the average yarn diameter. This is a low strength region in yarn and identifies as defect. From the test data it is clearly visible that ring yarn has lowest no of thin place. In 20 Ne there is none and in 30 Ne only 1 thin place is found. But in vortex yarn this number is higher. In 20 Ne there is no thin place but in 30 Ne yarn 16 thin places is found which is not so good.

Thick (+50%):

This refers to the area of the yarn where yarn diameter increases more than half of the average yarn diameter. This is a bulk region in yarn and identifies as defect it can cause shade variation in fabric. From the test data it is clearly visible that ring yarn has lowest no of thick place. In 20 Ne there is 4 and in 30 Ne 26 thick place is found. But in vortex yarn this number is higher. In 20 Ne there is 8 thick place but in 30 Ne yarn 34 thick places is found which is not so good as ring yarn.

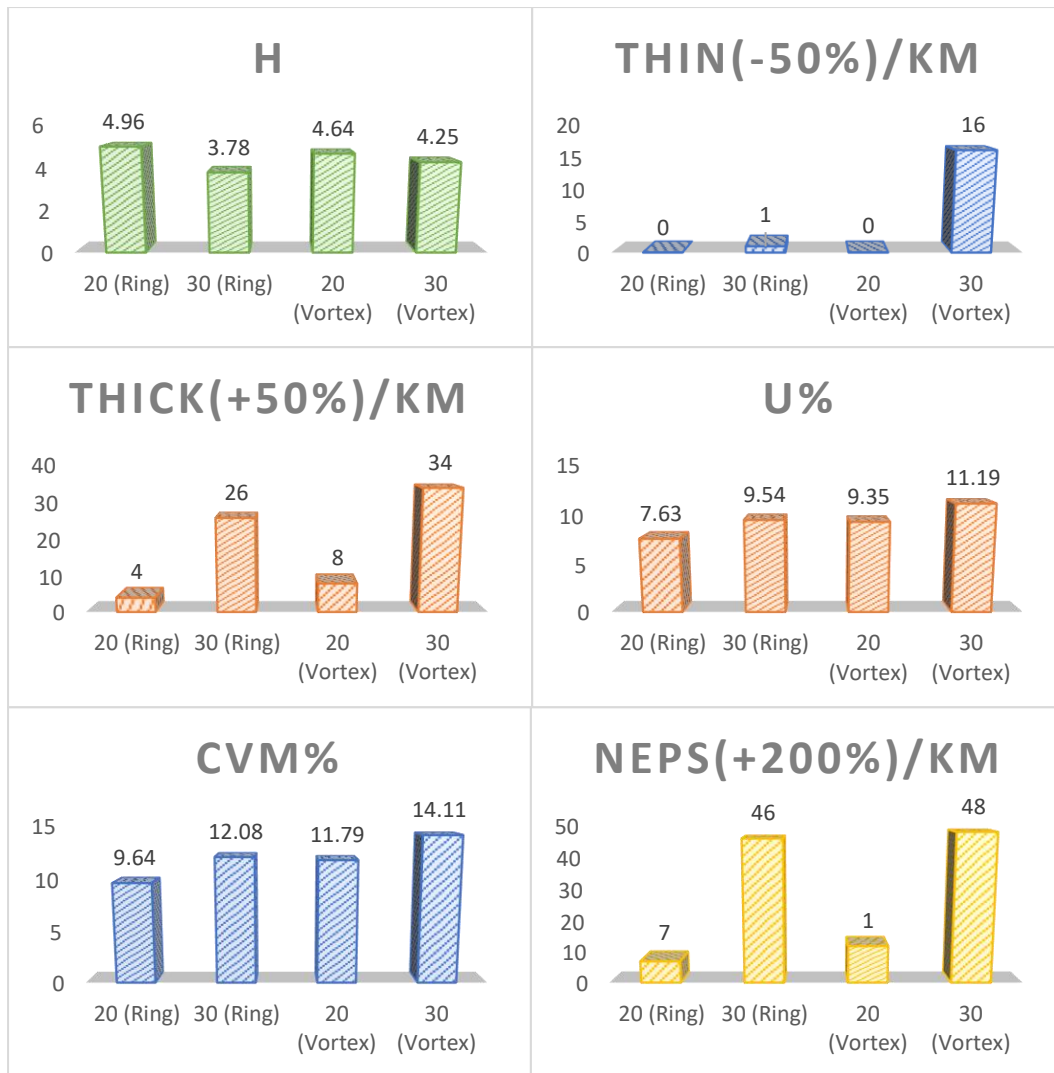


Figure 20: Quality comparison of 4 yarns

5.4. Test of Yarn Traction with MesdanLab Strength Tester

Count	Force [lb]	Tenacity [cN/tex]	CSP [Ne*lb]	Elongation [%]
20 (Ring)	188.165	2749.598	3650.395	11.859
30 (Ring)	105.602	2326.62	3088.844	10.172
20 (Vortex)	154.875	2275.976	3021.609	11.32
30 (Vortex)	89.618	2006.196	2663.446	10.008

CSP:

CSP stands for Count Strength Product. It is a measure that combines the count of the yarn with its strength, providing insights into the overall performance and suitability of

the yarn for specific applications. From the testing result of tri blend yarn, the CSP of ring yarn is 15-20% higher than the vortex yarn.

Tenacity:

Tenacity refers to the strength and durability of fibers and yarn, reflecting their ability to withstand tension, stress, and wear without breaking. It is a crucial property that influences the performance and longevity of yarn. By analysing the table data tenacity of ring yarn is higher than vortex yarn. In 20 Ne ring it is 2749.5 CN/tex and in 20 Ne vortex it is 2275.9 CN/tex. In 30 Ne ring it is 2326.6 CN/tex and in 30 Ne vortex it is 2006.19 CN/tex.

Elongation %:

Elongation is another significant property in yarn. It refers to the increase in length of yarn before break expressed in percentage. According to the table data both ring and vortex yarn shows almost identical elongation performance.

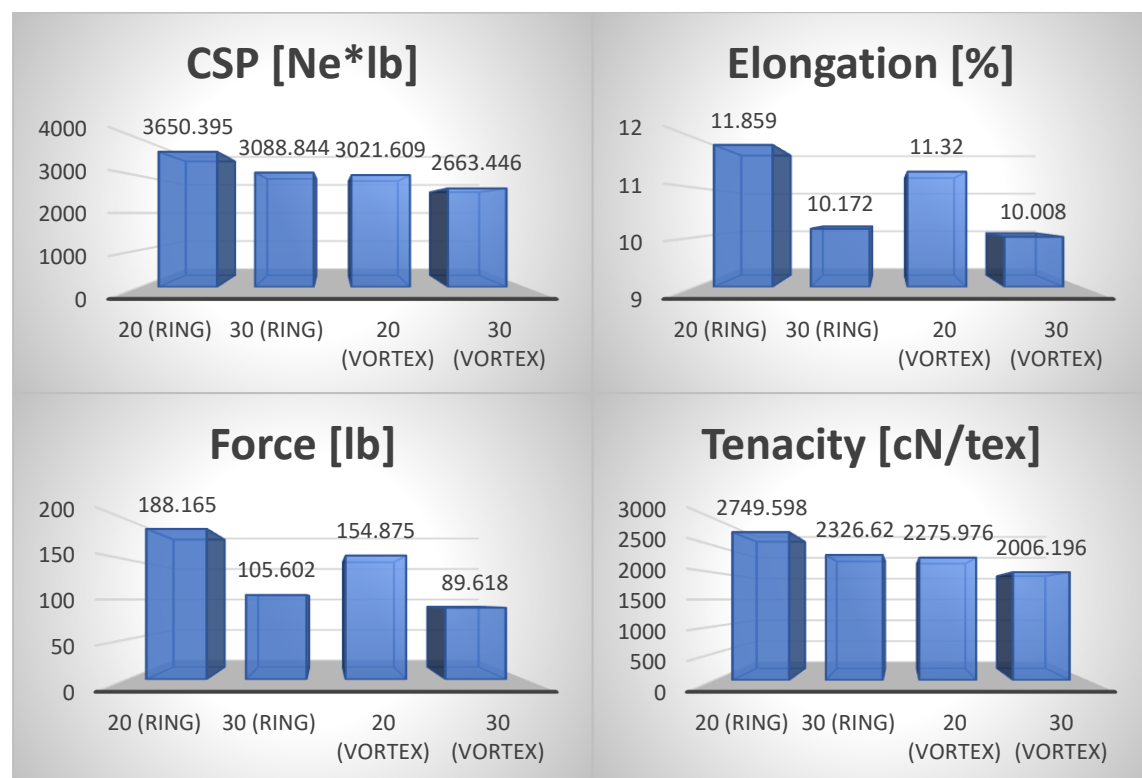


Figure 21: Mechanical properties comparison of 4 yarns

6. Limitations

- We could not ensure the same fiber length of cotton, modal and polyester fiber.
- Whether the percentage of cotton, modal and polyester fiber is correct is not tested.
- Proper Conditioning of cotton and modal fiber is not maintained.
- Produced yarn cannot be dyed by a single dye as it is a tri-blend. Cotton and Modal are cellulosic. They can be dyed by reactive dye. And Polyester can be dyed by disperse dye. But two processes have to be done separately. If we dye cotton first, when we will apply heat and chemical for polyester dyeing, cotton color will be degraded. So, only pigment dyeing is possible here. We can also produce mélange effect by dyeing only one fiber.

7. Conclusion

Recycled cotton, Polyester and Modal fibers have a great potential to be used as sustainable textile fiber for the apparel use in textile industries. Blend yarns of polyester and cellulosic fibers have the particular advantage of successfully combining the good properties of both fiber components, such as comfort of wear with easy-care properties. In the current work, the ratio of constituent fiber in the composition of the blend was found to be markedly influenced the characteristics of blend yarns.

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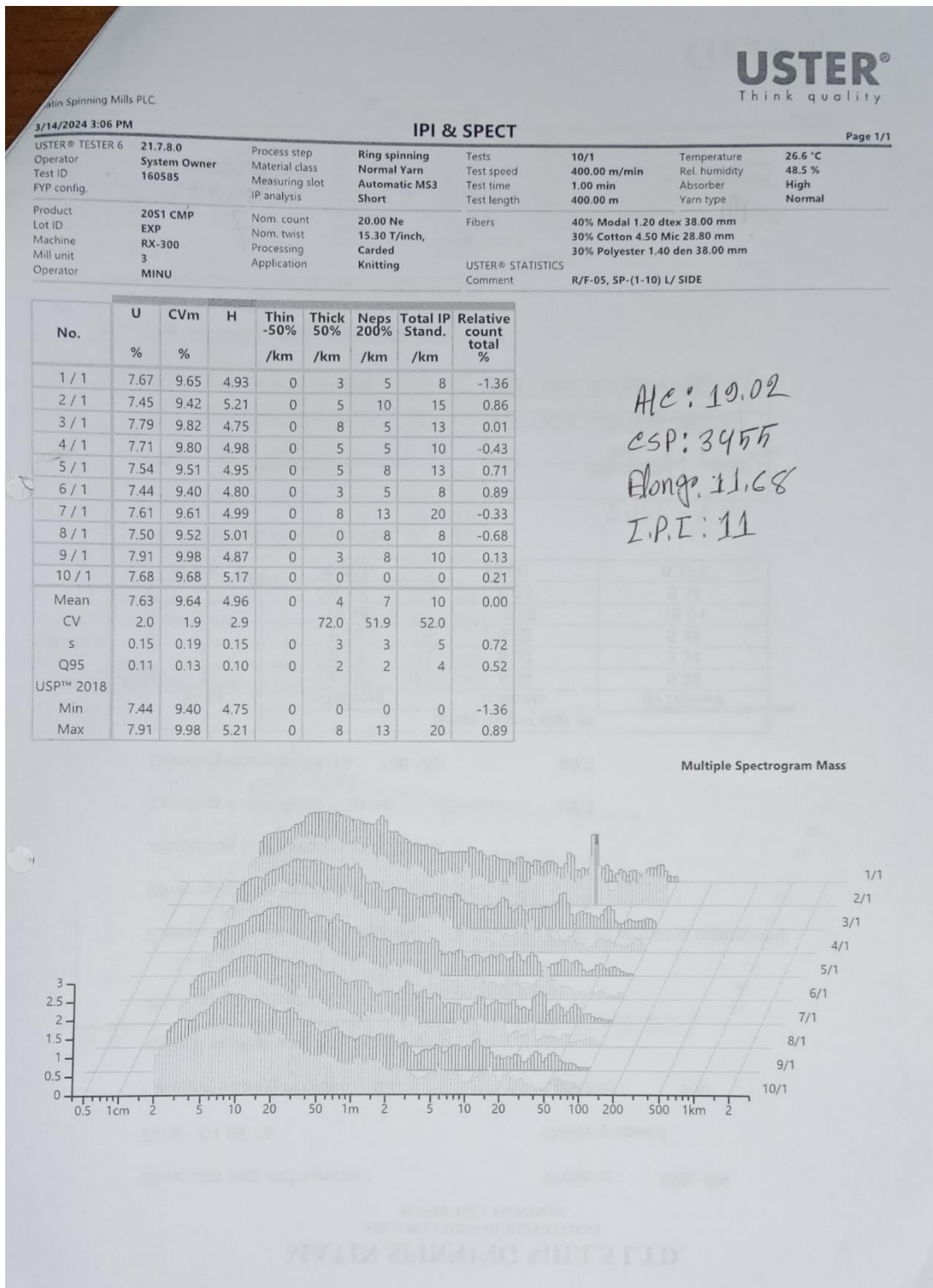
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9. Appendix



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Standard Table Yarn Report

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USTER® TESTER 6	21.7.8.0	Process step	Ring spinning	Tests	10/1	Temperature	26.6 °C
Operator	System Owner	Material class	Normal Yarn	Test speed	400.00 m/min	Rel. humidity	48.5 %
Test ID	160585	Measuring slot	Automatic MS3	Test time	1.00 min	Absorber	High
FYP config.		IP analysis	Short	Test length	400.00 m	Yarn type	Normal
Product	2051 CMP	Nom. count	20.00 Ne	Fibers	40% Modal 1.20 dtex 38.00 mm		
Lot ID	EXP	Nom. twist	15.30 T/inch		30% Cotton 4.50 Mic 28.80 mm		
Machine	RX-300	Processing	Carded		30% Polyester 1.40 den 38.00 mm		
Mill unit	3	Application	Knitting	USTER® STATISTICS			
Operator	MINU			Comment	R/F-05, 5P-(1-10) L/ SIDE		

No.	U	CVm	CVm 1m	CVm 3m	CVm 10m	Thin -30%	Thin -40%	Thin -50%	Thick 35%	Thick 50%	Neps 140%	Neps 200%	Neps 280%	Index	Relative count total %	H
	%	%	%	%	%	/km	/km	/km	/km	/km	/km	/km	/km			
1 / 1	7.67	9.65	3.03	2.28	1.47	95	0	0	40	3	23	5	3	1.36	-1.36	4.93
2 / 1	7.45	9.42	2.92	2.12	1.40	63	0	0	35	5	28	10	3	1.33	0.86	5.21
3 / 1	7.79	9.82	3.35	2.63	1.91	75	0	0	50	8	20	5	5	1.39	0.01	4.75
4 / 1	7.71	9.80	3.09	2.31	1.47	120	0	0	20	5	28	5	5	1.39	-0.43	4.98
5 / 1	7.54	9.51	2.79	2.11	1.51	98	0	0	38	5	35	8	3	1.34	0.71	4.95
6 / 1	7.44	9.40	2.76	2.18	1.56	90	0	0	38	3	20	5	3	1.33	0.89	4.80
7 / 1	7.61	9.61	3.06	2.25	1.60	115	3	0	50	8	28	13	5	1.36	-0.33	4.99
8 / 1	7.50	9.52	3.05	2.26	1.62	80	0	0	30	0	25	8	0	1.35	-0.68	5.01
9 / 1	7.91	9.98	3.02	2.27	1.51	180	0	0	48	3	28	8	0	1.41	0.13	4.87
10 / 1	7.68	9.68	2.88	2.19	1.65	73	3	0	43	0	33	0	0	1.37	0.21	5.17
Mean	7.63	9.64	2.99	2.26	1.57	99	1	0	39	4	27	7	3	1.36	0.00	4.96
CV	2.0	1.9	5.6	6.5	9.0	34.3	210.8		24.0	72.0	18.4	51.9	81.6	1.9		2.9
s	0.15	0.19	0.17	0.15	0.14	34	1	0	9	3	5	3	2	0.03	0.72	0.15
Q95	0.11	0.13	0.12	0.10	0.10	24	1	0	7	2	3	2	1	0.02	0.52	0.10
USP™ 2018																
Min	7.44	9.40	2.76	2.11	1.40	63	0	0	20	0	20	0	0	1.33	-1.36	4.75
Max	7.91	9.98	3.35	2.63	1.91	180	3	0	50	8	35	13	5	1.41	0.89	5.21

No.	sh	S3u	S1+2u
		/100m	/100m
1 / 1	0.99	1,995	13,238
2 / 1	1.02	2,320	14,808
3 / 1	0.94	1,669	11,410
4 / 1	0.99	2,052	12,647
5 / 1	0.97	1,657	12,774
6 / 1	0.97	1,940	12,433
7 / 1	1.02	2,383	13,202
8 / 1	1.00	2,207	14,664
9 / 1	0.97	1,907	11,201
10 / 1	1.07	2,482	13,230
Mean	0.99	2,061	12,961
CV	3.5	13.8	9.1
s	0.03	285	1,175
Q95	0.02	204	841
USP™ 2018			
Min	0.94	1,657	11,201
Max	1.07	2,482	14,808

Spinning Mills PLC

Standard Table Yarn Report

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USTER® TESTER 6	21.7.8.0	Process step	OE Rotor spinn	Tests	5/1	Temperature	26.4 °C
Operator	Operator	Material class	Normal Yarn	Test speed	400.00 m/min	Rel. humidity	57.8 %
Test ID	160721	Measuring slot	Automatic MS4	Test time	0.50 min	Absorber	Medium
FYP config.		IP analysis	Short	Test length	200.00 m	Yarn type	Normal
Product	30 CMP	Nom. count	30.00 Ne	Fibers	40% Modal 1.40 dtex 38.00 mm		
Lot ID	SAMPLE	Nom. twist	0.00 T/inch, Z		30% Cotton 4.50 Mic 28.80 mm		
Machine	ROTOR-04	Processing	Carded		30% Polyester 1.40 dtex 38.00 mm		
Mill unit	2	Application	Knitting	USTER® STATISTICS	100% CO, rotor yarn, carded, package, knitting, 2018		
Operator	MEHEDI			Comment	30 CMP VORTEX		

No.	U	CVm	CVm 1m	CVm 3m	CVm 10m	Thin -30%	Thin -40%	Thin -50%	Thick 35%	Thick 50%	Neps 140%	Neps 200%	Neps 280%	Index	Relative count total %	H
	%	%	%	%	%	/km	/km	/km	/km	/km	/km	/km	/km			
1 / 1	11.05	13.94	3.63	2.54	1.40	2,105	330	20	375	20	370	45	5	1.59	0.60	4.40
2 / 1	11.07	13.96	3.61	2.37	1.53	2,130	240	15	340	15	405	25	5	1.59	0.54	4.23
3 / 1	11.21	14.09	3.97	2.89	1.89	2,045	210	10	495	60	420	60	10	1.61	0.40	4.21
4 / 1	11.18	14.13	3.76	2.43	1.67	2,125	300	20	420	35	425	45	10	1.61	-0.38	4.23
5 / 1	11.43	14.42	4.06	2.81	1.68	2,410	335	15	480	40	520	65	10	1.65	-1.15	4.20
Mean	11.19	14.11	3.81	2.61	1.63	2,163	283	16	422	34	428	48	8	1.61	0.00	4.25
CV	1.4	1.4	5.3	8.8	11.1	6.6	19.7	26.1	15.7	52.4	13.0	32.6	34.2	1.4		1.9
s	0.15	0.19	0.20	0.23	0.18	142	56	4	66	18	56	16	3	0.02	0.75	0.08
Q95	0.19	0.24	0.25	0.29	0.23	176	69	5	82	22	69	19	3	0.03	0.94	0.10
USP™ 2018		13	21	8	<5	<5	23	6	<5	<5	<5	<5	<5			27
Min	11.05	13.94	3.61	2.37	1.40	2,045	210	10	340	15	370	25	5	1.59	-1.15	4.20
Max	11.43	14.42	4.06	2.89	1.89	2,410	335	20	495	60	520	65	10	1.65	0.60	4.40

No.	sh	S3u	S1+2u
		/100m	/100m
1 / 1	0.93	189	4,615
2 / 1	0.97	157	4,567
3 / 1	0.99	238	4,371
4 / 1	1.01	194	4,252
5 / 1	1.01	121	4,370
Mean	0.98	180	4,435
CV	3.5	24.3	3.4
s	0.03	44	151
Q95	0.04	54	188
USP™ 2018	<5	<5	<5
Min	0.93	121	4,252
Max	1.01	238	4,615

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Standard Table Yarn Report

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USTER® TESTER 6	21.7.8.0	Process step	OE Rotor spinn	Tests	5/1	Temperature	26.4 °C
Operator	Operator	Material class	Normal Yarn	Test speed	400.00 m/min	Rel. humidity	58.0 %
Test ID	160722	Measuring slot	Automatic M54	Test time	0.50 min	Absorber	Medium
FYP config		IP analysis	Short	Test length	200.00 m	Yarn type	Normal
Product	20 CMP VORTEX	Nom. count	30.00 Ne	Fibers	40% Modal 1.40 dtex 38.00 mm		
Lot ID	SAMPLE	Nom. twist	0.00 T/inch, Z		30% Cotton 4.50 Mic 28.80 mm		
Machine	ROTOR-04	Processing	Carded		30% Polyester 1.40 dtex 38.00 mm		
Mill unit	2	Application	Knitting	USTER® STATISTICS	100% CO, rotor yarn, carded, package, knitting, 2018		
Operator	MEHEDI			Comment	30 CMP VORTEX		

No.	U	CVm	CVm 1m	CVm 3m	CVm 10m	Thin -30%	Thin -40%	Thin -50%	Thick 35%	Thick 50%	Neps 140%	Neps 200%	Neps 280%	Index	Relative count total %	H
	%	%	%	%	%	/km	/km	/km	/km	/km	/km	/km	/km			
1 / 1	9.19	11.52	3.28	2.59	1.82	575	10	0	90	5	105	5	0	1.32	-0.76	4.73
2 / 1	9.24	11.67	2.99	2.33	1.31	735	35	0	110	10	150	5	0	1.33	-0.17	4.65
3 / 1	9.42	11.87	3.01	2.12	1.41	760	25	0	135	10	185	30	5	1.35	-0.22	4.63
4 / 1	9.52	12.02	3.17	2.31	1.53	835	20	0	150	10	150	5	5	1.37	-0.40	4.59
5 / 1	9.36	11.84	3.13	2.20	1.37	835	20	0	175	5	175	15	0	1.35	1.54	4.61
Mean	9.35	11.79	3.12	2.31	1.49	748	22	0	132	8	153	12	2	1.34	0.00	4.64
CV	1.4	1.6	3.8	7.8	13.4	14.2	41.3		25.2	34.2	20.2	91.3	136.9	1.6		1.2
s	0.13	0.19	0.12	0.18	0.20	107	9	0	33	3	31	11	3	0.02	0.89	0.05
Q95	0.17	0.24	0.15	0.22	0.25	132	11	0	41	3	38	14	3	0.03	1.11	0.07
USP™ 2018		<5	<5	<5	<5		<5	<5	<5	<5		<5	<5			40
Min	9.19	11.52	2.99	2.12	1.31	575	10	0	90	5	105	5	0	1.32	-0.76	4.59
Max	9.52	12.02	3.28	2.59	1.82	835	35	0	175	10	185	30	5	1.37	1.54	4.73

No.	sh	S3u	S1+2u
		/100m	/100m
1 / 1	0.95	137	3,862
2 / 1	1.01	66	3,770
3 / 1	1.04	59	3,939
4 / 1	1.03	94	3,620
5 / 1	1.04	75	3,669
Mean	1.02	86	3,772
CV	3.8	36.7	3.5
s	0.04	32	132
Q95	0.05	39	164
USP™ 2018	<5	<5	<5
Min	0.95	59	3,620
Max	1.04	137	3,939

20 Ne - Ring CMP (30+40+20)

Test of Yarn Traction with MesdanLab Strength Tester

Customer Code	MSY	Test Code	W-01	Date	3/16/2024 Time 2:10:48 PM
Sample Description					Sample Number 2
Parcel	Lot				Count 19.4 [Ne]
Material					
Examiner					
Sample Length	640 [mm]	Load cell ID/FS [kg]	5 / 500	X 1	
Clamp Speed	500 [mm/min]	Pretension	0 [lb]		
Test Machine					
Observations					

Statistical Results of the Test

	Force [lb]	Tenacity [cN/tex]	CLSP [Ne*lb]	Average Time of Breakage 9.1 [s]	Elongation [%]
Maximum	188.716 (1)	2757.652 (1)	3661.087		11.953 (2)
Minimum	187.614 (2)	2741.544 (2)	3639.703		11.766 (1)
Average	188.165	2749.598	3650.395		11.859
Range	0.586 [%]	0.586 [%]	0.586 [%]		1.581 [%]
CV	0.414 [%]	0.414 [%]	0.414 [%]		1.118 [%]
Deviation	0.779	11.390	15.121		0.133
IC95%	1.080	15.785	1.080		0.184
IC99%	1.420	20.745	1.420		0.241

Results of Single Samples

Sample #	Max Force [lb]	Time [s]	Tenacity [cN/tex]	CLSP [Ne*lb]	Max Elongation [%]
1	188.7	9.0	2757.652	3661.087	11.77
2	187.6	9.2	2741.544	3639.703	11.95

20 Ne - Vortex CMP (30+40+30)

Test of Yarn Traction with MesdanLab Strength Tester

Customer Code	MSY	Test Code	W-04	Date	3/16/2024 Time 2:27:21 PM
Sample Description					Sample Number 2
Parcel	Lot				Count 19.51 [Ne]
Material					
Examiner					
Sample Length	640 [mm]	Load cell ID/FS [kg]	5 / 500	X 1	
Clamp Speed	500 [mm/min]	Pretension	0 [lb]		
Test Machine					
Observations					

Statistical Results of the Test

	Force [lb]	Tenacity [cN/tex]	CLSP [Ne*lb]	Average Time of Breakage 8.7 [s]	Elongation [%]
Maximum	156.528 (2)	2300.274 (2)	3053.868		11.438 (2)
Minimum	153.221 (1)	2251.677 (1)	2989.349		11.203 (1)
Average	154.875	2275.976	3021.609		11.320
Range	2.135 [%]	2.135 [%]	2.135 [%]		2.07 [%]
CV	1.510 [%]	1.510 [%]	1.510 [%]		1.464 [%]
Deviation	2.338	34.364	45.621		0.166
IC95%	3.241	47.624	3.241		0.230
IC99%	4.259	62.589	4.259		0.302

Results of Single Samples

Sample #	Max Force [lb]	Time [s]	Tenacity [cN/tex]	CLSP [Ne*lb]	Max Elongation [%]
1	153.2	8.6	2251.677	2989.349	11.20
2	156.5	8.8	2300.274	3053.868	11.44

30 Ne — Vortex CMP (30+40+30)

Test of Yarn Traction with MesdanLab Strength Tester

Customer Code	MSY	Test Code	W-03	Date	3/16/2024 Time 2:22:20 PM	
Sample Description					Sample Number	2
Parcel	Lot				Count	29.72 [Ne]
Material						
Examiner						
Sample Length	640 [mm]	Load cell ID/FS [kg]		5 / 500	X 1	
Clamp Speed	500 [mm/min]	Pretension		0 [lb]		
Test Machine						
Observations						

Statistical Results of the Test

	Force [lb]	Tenacity [cN/tex]	CLSP [Ne*lb]	Average Time of Breakage [s]	7.7 [s]
Maximum	91.271 (2)	2043.211 (2)	2712.588	10.219 (2)	
Minimum	87.965 (1)	1969.181 (1)	2614.305	9.797 (1)	
Average	89.618	2006.196	2663.446	10.008	
Range	3.690 [%]	3.690 [%]	3.690 [%]	4.215 [%]	
CV	2.609 [%]	2.609 [%]	2.609 [%]	2.981 [%]	
Deviation	2.338	52.347	69.496	0.298	
IC95%	3.241	72.547	3.241	0.413	
IC99%	4.259	95.344	4.259	0.543	

Results of Single Samples

Sample #	Max Force [lb]	Time [s]	Tenacity [cN/tex]	CLSP [Ne*lb]	Max Elongation [%]
1	88.0	7.5	1969.181	2614.305	9.80
2	91.3	7.8	2043.211	2712.588	10.22

30 Ne — Ring CMP (30+40+30)

Test of Yarn Traction with MesdanLab Strength Tester

Customer Code	MSY	Test Code	W-02	Date	3/16/2024 Time 2:16:22 PM	
Sample Description					Sample Number	2
Parcel	Lot				Count	29.25 [Ne]
Material						
Examiner						
Sample Length	640 [mm]	Load cell ID/FS [kg]		5 / 500	X 1	
Clamp Speed	500 [mm/min]	Pretension		0 [lb]		
Test Machine						
Observations						

Statistical Results of the Test

	Force [lb]	Tenacity [cN/tex]	CLSP [Ne*lb]	Average Time of Breakage [s]	7.8 [s]
Maximum	106.042 (2)	2336.334 (2)	3101.741	10.500 (1)	
Minimum	105.161 (1)	2316.905 (1)	3075.947	9.844 (2)	
Average	105.602	2326.620	3088.844	10.172	
Range	0.835 [%]	0.835 [%]	0.835 [%]	6.452 [%]	
CV	0.590 [%]	0.590 [%]	0.590 [%]	4.562 [%]	
Deviation	0.624	13.738	18.239	0.464	
IC95%	0.864	19.040	0.864	0.643	
IC99%	1.136	25.023	1.136	0.845	

Results of Single Samples

Sample #	Max Force [lb]	Time [s]	Tenacity [cN/tex]	CLSP [Ne*lb]	Max Elongation [%]
1	105.2	8.1	2316.905	3075.947	10.50
2	106.0	7.6	2336.334	3101.741	9.84

Summary Table

USTER® AFIS PRO 2

Company Uster Technologies, Inc.
 Sample ID 30.990
 Sample Type FINISHER SLIVER
 Program ID FINISHER SLIVER
 Article MSML
 Blend Level 100% COTTON
 Machine FINISHER
 Machine Nr 2
 Operator ID MEHEDI
 Print Date & Time 3/10/2024 3:37:10PM
 Test Date & Time 3/10/2024 10:46:55AM
 AFIS SW version 1.4.4.140
 AFIS Serial Nr 2106533
 Report ID 1.0
 USTER® STATISTICS None
 Location QUALITY CONTROL
 Comment CMP(30+40+30)
 Rel Humidity 42.7%
 Temp 26.8 C

NEP	Rep	Total Nep Cnt		Total Nep Mean Size		Fiber Nep Cnt		FibNep Mean Size		SCNep Count		SCNep Mean Size									
		[Cnt / g]		[µm]		[Cnt / g]		[µm]		[Cnt / g]		[µm]									
Mean Std. Dev. CV% Q99% Min Max USP	1	21	547	21	547	0	547	0	547	0	547	0									
	2	21	535	21	535	0	535	0	535	0	535	0									
	3	32	619	32	619	0	619	0	619	0	619	0									
	n	25	567	25	567	0	567	0	567	0	567	0									
	Std. Dev.	6	45	6	45	0	45	0	45	0	45	0									
	CV%	25.7	8.0	25.7	8.0	0.0	8.0	0.0	8.0	0.0	8.0	0.0									
	Q99%	34	258	34	258	0	258	0	258	0	258	0									
	Min	21	535	21	535	0	535	0	535	0	535	0									
	Max	32	619	32	619	0	619	0	619	0	619	0									
	USP	[---]																			
LEN	Rep	L(w)		SFC(w)		UOL(w)		L(n)		SFC(n)		5% L(n)		Fineness		Maturity		IFC			
		[mm]	[%CV]	%<12.7 mm	[mm]	[mm]	[%CV]	%<12.7 mm	[mm]	[mm]	[mtex]	Ratio [mat]	[%]								
Mean Std. Dev. CV% Q99% Min Max USP	1	34.8	22.8	1.3	39.2	31.7	31.5	5.3	42.0	165	1.05	3.1	165	1.05	3.1	1.06	3.4	1.06	3.4		
	2	34.8	23.2	1.6	39.4	31.4	32.6	6.5	42.2	163	1.06	3.2	163	1.06	3.2	1.05	3.1	1.05	3.1		
	3	35.1	23.5	1.6	39.4	31.6	33.3	6.7	42.4	167	1.06	4.0	167	1.06	4.0	1.06	4.0	1.06	4.0		
	n	3	34.9	23.2	1.5	39.3	31.6	32.5	6.2	42.2	165	1.06	3.4	165	1.06	3.4	1.06	3.4	1.06	3.4	
	Std. Dev.	0.2	0.3	0.2	0.1	0.1	0.9	0.8	0.2	2	0.00	0.5	2	0.00	0.5	0.00	0.00	0.5	0.00	0.5	
	CV%	0.4	1.4	12.7	0.3	0.4	2.9	12.6	0.4	1.2	0.4	14.4	1.2	0.4	14.4	0.03	0.03	14.4	0.03	14.4	
	Q99%	0.9	1.9	1.1	0.7	0.7	5.4	4.4	1.0	11	0.03	2.8	11	0.03	2.8	0.03	0.03	2.8	0.03	2.8	
	Min	34.8	22.8	1.3	39.2	31.4	31.5	5.3	42.0	163	1.05	3.1	163	1.05	3.1	1.05	3.1	1.05	3.1	1.05	3.1
	Max	35.1	23.5	1.6	39.4	31.7	33.3	6.7	42.4	167	1.06	4.0	167	1.06	4.0	1.06	4.0	1.06	4.0	1.06	4.0
	USP	[---]																			

Company Uster Technologies, Inc.				AFIS SW version 1.4.4.140								
Sample ID 30.996				AFIS Serial Nr 2106533								
Sample Type BREAKER SLIVER				Report ID 1.0								
Program ID BREAKER SLIVER				USTER® STATISTICS None								
Article MSML				Location QUALITY CONTROL								
Blend Level CMP(30+40=30)				Comment B/D:2nd, CMP(30+40=30)								
Machine BREAKER				Rel Humidity 42.1%								
Machine Nr 3				Temp 27.4 C								
Operator ID MEHEDI				Print Date & Time 3/16/2024 1:30:24PM								
Test Date & Time 3/16/2024 1:30:24PM				Test Date & Time 3/10/2024 1:59:53PM								
NEP	Rep	Total Nep Cnt	Total Nep Mean Size	Fiber Nep Cnt	FibNep Mean Size	SCNep Count	SCNep Mean Size					
		[Cnt / g]	[µm]	[Cnt / g]	[µm]	[Cnt / g]	[µm]					
	1	34	598	32	607	2	475					
	2	19	545	19	545	0	0					
	3	28	554	28	554	0	0					
	n											
	Mean	27	566	26	569	1	475					
	Std. Dev.	8	28	7	34	1	0					
	CV%	28.0	5.0	25.3	5.9	173.2	0.0					
	Q99%	46	160	40	195	6	0					
	Min	19	545	19	545	0	0					
	Max	34	598	32	607	2	475					
	USP	[---]										
	LEN	Rep	L(w)	L(w) CV%	SFC(w)	UOL(w)	L(n)	L(n) CV%	SFC(n)	5% L(n)	Fineness	Maturity
		[mm]	[%CV]	%<12.7 mm	[mm]	[mm]	[%CV]	%<12.7 mm	[mm]	[mtext]	[ratio]	[%]
	1	33.1	23.6	1.8	37.8	29.6	34.1	7.6	40.5	158	1.03	4.1
	2	33.1	25.0	2.1	37.8	29.5	35.2	8.2	40.7	160	1.02	4.4
	3	33.1	25.3	2.3	37.9	29.2	36.7	9.6	40.9	161	1.03	4.0
	n											
	Mean	33.1	24.6	2.1	37.9	29.4	35.3	8.5	40.7	160	1.03	4.2
	Std. Dev.	0.0	0.9	0.2	0.0	0.2	1.3	1.0	0.2	2	0.00	0.2
	CV%	0.1	3.8	11.4	0.1	0.8	3.6	11.9	0.5	1.0	0.5	4.4
	Q99%	0.2	5.4	1.3	0.2	1.4	7.3	5.8	1.2	11	0.03	1.1
	Min	33.1	23.6	1.8	37.8	29.2	34.1	7.6	40.5	158	1.02	4.0
	Max	33.1	25.3	2.3	37.9	29.6	36.7	9.6	40.9	161	1.03	4.4
	USP	[---]										

3/16/2024 1:30:24PM

Page 1 of 1

Summary Table

USTER® AFIS PRO 2

Company Uster Technologies, Inc.

Sample ID 30.991

AFIS SW version 1.4.4.140

Sample Type BREAKER SLIVER

AFIS Serial Nr 2106533

Program ID BREAKER SLIVER

USTER®

Report ID 1.0

Article MSML

Blend Level CMP(30+40=30)

Location QUALITY CONTROL

Machine BREAKER SLIVER

Comment CMP(30+40=30)

Machine Nr 3

Rel Humidity 43.1%

Operator ID MEHEDI

Temp 26.7 C

Print Date & Time 3/10/2024 3:37:00PM

Test Date & Time 3/10/2024 10:55:03AM

NEP	Rep	Total Nep Cnt		Total Nep Mean Size		Fiber Nep Cnt		FibNep Mean Size		SCNep Count		SCNep Mean Size											
		[Cnt / g]		[µm]		[Cnt / g]		[µm]		[Cnt / g]		[µm]											
LEN	1	38	572	38	572	0	572	0	572	0	572	0	0										
	2	33	540	33	540	0	540	0	540	0	540	0	0										
	3	22	580	22	580	0	580	0	580	0	580	0	0										
	n	31	564	31	564	0	564	0	564	0	564	0	0										
	Mean	8	21	8	21	0	21	0	21	0	21	0	0										
	Std. Dev.	26.4	3.8	26.4	3.8	0.0	3.8	0.0	3.8	0.0	3.8	0.0	0.0										
	CV%	46	120	46	120	0	120	0	120	0	120	0	0										
	Q99%	22	540	22	540	0	540	0	540	0	540	0	0										
	Min	38	580	38	580	0	580	0	580	0	580	0	0										
	Max	38	580	38	580	0	580	0	580	0	580	0	0										
USP	[—]																						
LEN	Rep	L(w)		L(w) CV%		SFC(w)		UQL(w)		L(n)		L(n) CV%		SFC(n)		5% L(n)		Fineness		Maturity		IFC	
		[mm]		[%CV]		[%<12.7 mm]		[mm]		[mm]		[%CV]		[%<12.7 mm]		[mm]		[mte]		[mat]		[%]	
	1	33.1	24.5	2.3	38.1	29.4	35.4	8.9	40.7	163	1.03	4.0	1.03	4.1	1.03	4.1	1.03	167	1.04	1.03	1.04	3.7	3.7
	2	33.3	23.6	2.3	38.0	29.8	34.1	8.6	40.7	167	1.04	4.1	1.04	4.1	1.04	4.1	1.04	167	1.04	1.03	1.03	4.1	4.1
	3	33.3	23.8	2.0	38.0	29.7	34.7	8.4	40.8	160	1.03	3.0	1.03	3.0	1.03	3.0	1.03	160	1.03	1.03	1.03	3.0	3.0
	n	33.2	24.0	2.2	38.0	29.7	34.7	8.6	40.7	163	1.04	3.7	1.04	3.7	1.04	3.7	1.04	163	1.04	1.04	1.04	3.7	3.7
	Mean	0.1	0.5	0.1	0.0	0.2	0.7	0.3	0.1	4	0.00	0.6	0.00	0.6	0.00	0.6	0.00	4	0.00	0.00	0.00	0.6	0.6
	Std. Dev.	0.3	2.0	6.9	0.1	0.7	1.9	3.2	0.2	2.2	0.4	17.2	0.4	17.2	0.4	17.2	0.4	2.2	0.4	0.02	0.02	3.6	3.6
	CV%	0.6	2.8	0.9	0.3	1.1	3.8	1.6	0.5	23	0.02	3.6	0.02	3.6	0.02	3.6	0.02	23	0.02	0.02	0.02	3.6	3.6
	Q99%	2.0	23.6	2.0	38.0	29.4	34.1	8.4	40.7	160	1.03	3.0	1.03	3.0	1.03	3.0	1.03	160	1.03	1.03	1.03	3.0	3.0
	Min	33.1	23.6	2.0	38.0	29.4	34.1	8.4	40.7	160	1.03	3.0	1.03	3.0	1.03	3.0	1.03	160	1.03	1.03	1.03	3.0	3.0
	Max	33.3	24.5	2.3	38.1	29.8	35.4	8.9	40.8	167	1.04	4.1	1.04	4.1	1.04	4.1	1.04	167	1.04	1.04	1.04	4.1	4.1
USP	[—]																						

Summary Table

USTER® AFIS PRO 2

Company Uster Technologies, Inc.

Sample ID 30.995

Sample Type CARDING SLIVER

Program ID CARD SLIVER

Article MSML

Blend Level CMP(30+40=30)

Machine CARDING

Machine Nr 3

Operator ID MEHEDI

Print Date & Time 3/10/2024 3:36:45PM

AFIS SW version 1.4.4.140

AFIS Serial Nr 2106533

Report ID 1.0

USTER® STATISTICS None

Location QUALITY CONTROL

Comment CMP(30+40=30)

Rel Humidity 43.3%

Temp 27.3 C

Test Date & Time 3/10/2024 12:48:04PM

NEP	Rep	Total Nep Cnt	Total Nep Mean Size	Fiber Nep Cnt	FibNep Mean Size	SCNep Count	SCNep Mean Size					
		[Cnt / g]	[µm]	[Cnt / g]	[µm]	[Cnt / g]	[µm]					
	1	36	581	36	581	0	0					
	2	29	548	29	548	0	0					
	3	37	549	37	549	0	0					
	n	3										
Mean		34	559	34	559	0	0					
Std. Dev.		4	19	4	19	0	0					
CV%		12.8	3.4	12.8	3.4	0.0	0.0					
Q99%		23	109	23	109	0	0					
Min		29	548	29	548	0	0					
Max		37	581	37	581	0	0					
USP												
	[—]											
LEN	Rep	L(w)	L(w) CV%	SFC(w)	UOL(w)	L(n)	L(n) CV%	SFC(n)	5% L(n)	Fineness	Maturity Ratio	IFC
		[mm]	[%CV]	%<12.7 mm	[mm]	[mm]	[%CV]	%<12.7 mm	[mm]	[mm]	[mat]	[%]
	1	32.8	24.8	2.3	37.9	29.1	36.1	9.1	40.7	164	1.04	3.3
	2	33.2	23.6	1.6	37.8	29.9	33.0	6.7	40.5	159	1.02	4.2
	3	33.3	23.6	1.8	37.8	30.0	33.2	7.0	40.5	157	1.03	4.6
	n	3										
Mean		33.1	24.0	1.9	37.8	29.7	34.1	7.6	40.6	160	1.03	4.0
Std. Dev.		0.2	0.7	0.4	0.1	0.5	1.7	1.3	0.1	4	0.01	0.7
CV%		0.7	3.0	19.4	0.2	1.7	5.1	17.4	0.2	2.3	0.8	16.8
Q99%		1.3	4.1	2.1	0.5	3.0	9.9	7.6	0.5	2.3	0.05	3.9
Min		32.8	23.6	1.6	37.8	29.1	33.0	6.7	40.5	157	1.02	3.3
Max		33.3	24.8	2.3	37.9	30.0	36.1	9.1	40.7	164	1.04	4.6
USP												
	[—]											